

Structure Output Tasks

LangChain Practice Worksheet

■ Tasks for Structured Output

1. Student Management System

Create a LangChain prompt that takes a student's information and returns structured output.

Input example: My name is Rahul, I am 21 years old, studying Computer Science in 3rd year, and my percentage is 78.5.

Expected structure: name → string; age → integer; course → string; year → integer; percentage → number.

Task: Define a suitable Pydantic structure and make the LLM return the information according to the schema.

2. Patient Management System

Create a structured-output system that extracts patient information from a natural-language prompt.

Input example: Patient name is Priya Sharma, age 35, suffering from fever and headache. Her blood group is O+.

Expected structure: patient_name → string; age → integer; symptoms → list of strings; blood_group → string.

Task: Define a suitable Pydantic structure and make the LLM return the information according to the schema.

3. Library Book Management

Create a system that extracts book information from a user's message.

Input example: I want to issue the book Python Programming by Mark Lutz. It has 800 pages and belongs to the programming category.

Expected structure: book_name → string; author → string; pages → integer; category → string; available → boolean.

Task: Define a suitable Pydantic structure and make the LLM return the information according to the schema.

4. Job Applicant Management

Create a structured-output system that extracts job applicant information.

Input example: My name is Amit Patil. I have 2 years of experience in React and Node.js. I am expecting a salary of 6 LPA and can join immediately.

Expected structure: name → string; experience → integer; skills → list of strings; expected_salary → number; immediate_joiner → boolean.

Task: Define a suitable Pydantic structure and make the LLM return the information according to the schema.

■ Tasks for Output Parser

1. Student Result Parser

Create a prompt that asks the LLM to generate a student's result.

Input: Student Rahul has scored 85 in Python, 78 in JavaScript and 90 in DBMS.

Expected LLM output:

Name: Rahul
Python: 85
JavaScript: 78
DBMS: 90
Total: 253
Percentage: 84.33

Task: Use an output parser to parse the LLM response and extract each value separately.

2. Patient Information Parser

Ask the LLM to generate patient information in a fixed text format.

Input: Patient Ravi is 45 years old and has fever and cough.

Expected LLM output:

Name: Ravi
Age: 45
Symptoms: Fever, Cough

Task: Create an output parser that extracts Name, Age and Symptoms.

3. Library Book Parser

Create a prompt that generates book information in a predefined format.

Input: Harry Potter was written by J.K. Rowling. It has 500 pages and belongs to the Fantasy category.

Expected LLM output:

Book: Harry Potter
Author: J.K. Rowling
Pages: 500
Category: Fantasy

Task: Use an output parser to extract the four fields from the LLM response.

4. Job Candidate Parser

Create a prompt that generates candidate information in a fixed format.

Input: Sneha is a Python developer with 3 years of experience. She knows Python, Django and SQL and expects 7 LPA.

Expected LLM output:

Name: Sneha
Role: Python Developer
Experience: 3 years
Skills: Python, Django, SQL
Expected Salary: 7 LPA

Task: Use an output parser to extract the candidate information into separate variables.

■ Quick Concept Difference

Structured Output	Output Parser
Define the structure/schema	Define how to parse the response
Often uses Pydantic models	Can use StrOutputParser, CommaSeparatedListOutputParser, PydanticO
Useful for reliable data extraction	Useful for converting LLM responses into usable formats